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Tunable Mechanism Enables Robust Surface Perching Under Different Landing Impacts and Orientation Misalignment

Mahmud Hasan Saikot, Feiyu Wu, Elisha Lerner, Bo Cheng, and Jianguo Zhao**

Perching significantly enhances the energy efficiency and operational versatility of aerial robots. This article introduces a passive and tunable perching mechanism designed for smooth surfaces. The design features a bistable mechanism (BM) with a soft suction cup, augmented by two sets of shape memory alloy (SMA) actuators for active tuning. The BM enables rapid attachment upon surface contact. A set of SMA wires can increase the BM's triggering force to handle high contact speeds, while a set of SMA springs attached to the suction cup's edges can pull the cup to handle orientation misalignment. Experiments are conducted to characterize how the SMA actuators influence the BM's triggering force and the suction cup's displacement under continuous steady-state low-voltage heating. Additional experiments demonstrate fast tuning using momentary high-voltage heating of the SMA actuators to enhance energy efficiency. The mechanism enables successful perching on smooth surfaces and adapt to varying contact speeds and misalignments when properly tuned for three scenarios: pendulum-based perching, ground perching, and ceiling perching. With its tuning capability, the perching mechanism can alleviate the need for precise motion control for an aerial robot during perching, expanding the applications of aerial robots in areas like environmental monitoring or infrastructure inspection.

1. Introduction

In recent years, the usage of aerial robots has surged across a wide range of applications, including transportation, construction, precision agriculture, and 3D mapping, among others.[1,2] However, due to power constraints, most existing aerial robots, particularly multirotors, face significant challenges in sustained operations. The ability to perch or rest on objects offers a promising solution to this problem, enabling energy conservation during extended tasks, such as aerial monitoring.[3,4]

Perching can be categorized based on the objects that aerial robots can perch and the mechanism used to do so. One of the most common types involves perching with grasping or grabbing onto an object, such as a tree branch, mimicking birds, with some recent notable works.[5–9] Another major category encompasses flat surfaces like walls or ceilings. In particular, smooth surfaces represent an important category.[2] The ability to perch on smooth surfaces has significant potential to enhance the capabilities of aerial robots. For instance, it facilitates perching onto high-rise structures (e.g., wind turbines, skyscrapers),[10] enabling crucial inspections and maintenance. It allows for safe and secure landing onto moving objects such as vehicles navigating uneven terrain,[11,12] or ship decks amidst unpredictable waves at sea.[13,14]

Achieving successful perching on surfaces necessitates a combination of precise motion control (computational intelligence) and a reliable perching mechanism (mechanical intelligence). Computational intelligence involves sophisticated control algorithms that enable aerial robots to attain the desired orientation and speed before contacting the surface relying on real-time estimation and control.[12,15–17] However, recent advancements suggest that well-designed perching mechanisms, harnessing mechanical intelligence or the concept of physical intelligence,[18–21] can mitigate the need for precise motion control and reduce computational demands, enabling more efficient and robust perching capabilities.

Building on the concept of physical intelligence, researchers have investigated various mechanisms to tackle the challenge of perching onto surfaces. These mechanisms encompass dry or other adhesives, electrostatic forces, suction cups, and other methods. In the case of dry adhesives, several adhesive pads are usually affixed to an aerial robot that ensures the attachment onto the surface.[22–25] Other adhesives based on van der Waals force have also been utilized, such as biomimetic wet adhesive pads,[26] gecko-inspired directional adhesive,[27] and biomimetic adhesive material.[28] Electrostatic adhesion is typically achieved by applying a high voltage to some electrodes or pads on the robot, which generates an attractive force between the robot

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and the perching surface.[29–31] Suction cups typically utilize miniature vacuum pumps to create a strong attachment to the surface.[32–35] Some designs can also accommodate angular misalignments with the surface.[33–35] Suction-based grippers, such as the compliant self-sealing mechanism[36] can handle large misalignments and displacement errors. Alternatively, passive suction cups rely on impact forces for attachment.[37,38] Besides these, there are other methods such as microspines,[39] reverse thrust,[40] flip and flap,[41] etc.

In this article, we investigate how to enable robust perching onto smooth surfaces under different conditions by leveraging a perching mechanism with mechanical intelligence to overcome the challenges of precise control. Our design features a tunable mechanism with a suction cup attached to a bistable mechanism (BM) that can absorb different landing impacts and accommodate orientation misalignments while perching. The basic idea behind the tuning mechanism is illustrated in **Figure 1**. In some scenarios such as inverted landing, a quadrotor may not be able to control its speed or orientation anymore right before touchdown.[17] In such cases, instead of precisely controlling an aerial robot's speed, our mechanism can adjust the triggering force for a BM to absorb different landing impacts (Figure 1A), ensuring adaptability to a wide range of perching speeds. Furthermore, instead of precisely controlling an aerial robot's orientation, we can pull the suction cup to adjust its orientation to compensate for misalignments, enhancing the overall perching performance (Figure 1B). Through rigorous experimentation, we demonstrate the effectiveness of our approach in achieving robust perching on smooth surfaces.

The primary contribution of our study is to investigate how active tuning of the perching mechanism can enable robust perching. To our knowledge, generally no existing mechanism can adjust its properties to deal with various landing conditions for successful perching. This tuning capability enhances the robustness of the mechanism by alleviating the need for precise motion control. It accommodates different perching conditions, including different landing impacts and orientation misalignments, with our experimental results demonstrating functionality up to 30° of misalignment at contact speeds up to 2.16 m s^{-1} using a pendulum setup. Ground-based perching experiments further validated the effectiveness of tuning, achieving successful perching at misalignments of up to 40° and contact speeds reaching 1.98 m s^{-1}. Ceiling perching experiments showcased success at misalignments as high as 45°, with active drone thrust significantly enhancing perching success under challenging conditions. The mechanism presented in this article is based on our previous work,[38] where active tuning is not considered. In addition to the tuning mechanism, we have also enhanced our previous design by integrating a self-locking mechanism that can lock the whole mechanism upon successful touchdown on the surface. This self-locking mechanism further improves the reliability of the mechanism, particularly under higher landing impacts.

The rest of this article is organized as follows: in Section 2, we explain the working principle and the perching process with the tunable mechanism. Section 3 discusses the design and fabrication of the mechanism. In Section 4, the experimental results are presented and discussed. In Section 5, we summarize our work and discuss future works.

2. Working Principle and Perching Procedure

In this section, we introduce the working principle of our tunable perching mechanism. Furthermore, we detail the sequential process of perching with the mechanism step by step.

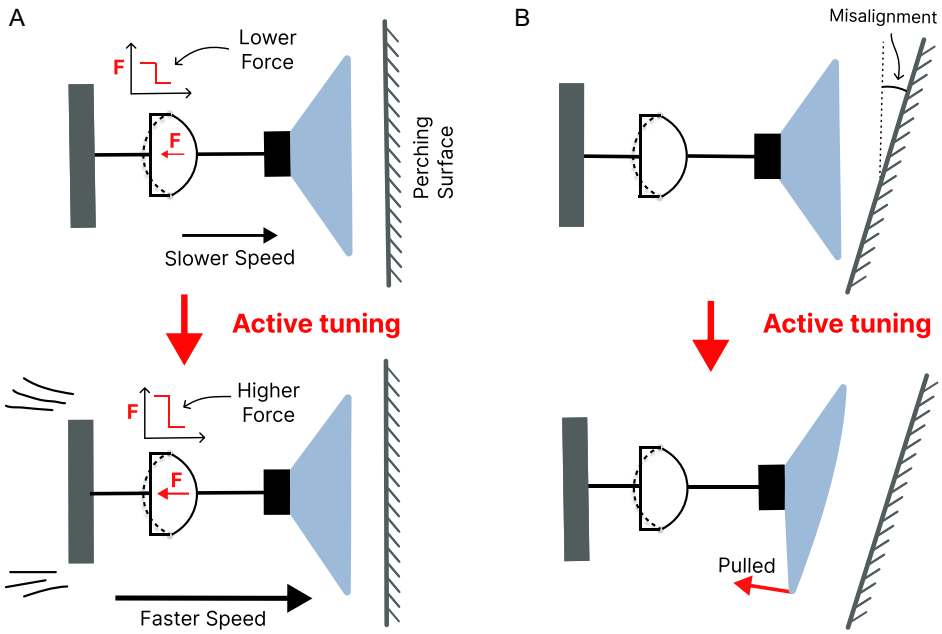


Figure 1. Illustration of our passive tuning mechanism. A) The mechanism can adapt to different contact speeds by actively tuning its triggering force for a bistable mechanism to absorb different impacts; B) the mechanism can adapt to orientation misalignments by pulling the edge of a suction cup.

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2.1. Working Principle

The tunable perching mechanism improves on our previous work,[38] and it includes four main parts (**Figure 2**D): a BM, a customized soft suction cup, a self-locking mechanism, and a tuning mechanism.

The BM (Figure 2A) enables swift perching by rapidly switching between two stable states. It is a beam-based design with two flexible beams attached to a rigid base in parallel. A rigid switch part in the middle is connected to the end of the two beams through a rigid link and two revolute joints on each side. With the mechanism initially at the extended stable state (left of Figure 2A), if an upward force is applied to the switch part, the two beams will bend outward, increasing the elastic energy stored in the beams to a maximum value at an intermediate unstable state. After passing the unstable state, the mechanism will automatically switch to the collapsed stable state (right of Figure 2A). In this state, if a downward force is applied to the switch part, it will switch back to the extended state. This force required to trigger the change of BM state is called the triggering

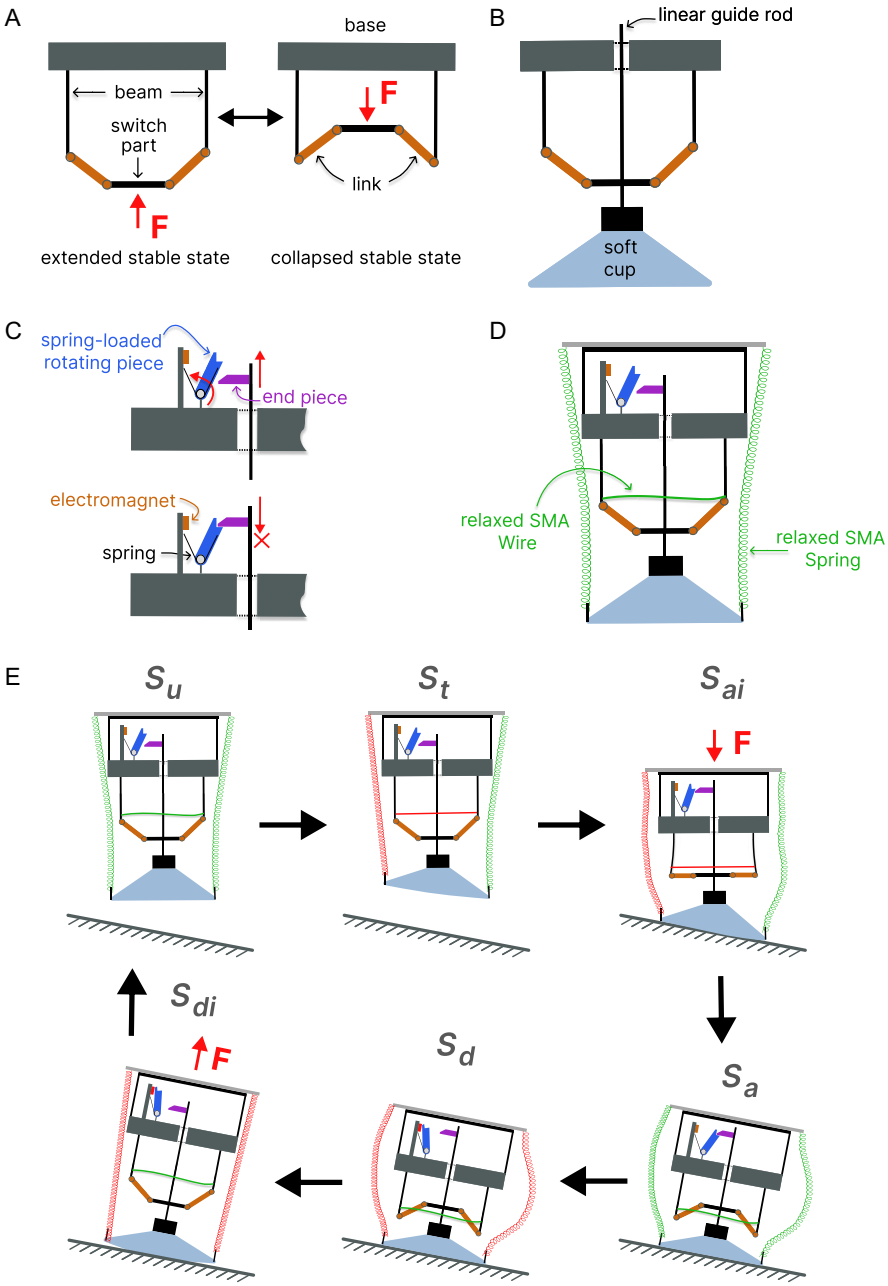


Figure 2. Working principle of the tunable perching mechanism. A) A beam-based bistable mechanism (BM); B) the BM with a customized suction cup; C) the self-locking mechanism; D) the tuning mechanism with SMA wires to tune the BM's triggering force and SMA springs to pull the soft cup; E) illustration of the perching process using the tunable mechanism.

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force (F_t). The detailed working principle can be found in our previous work.[21]

To enable perching onto flat and smooth surfaces, we create a customized soft suction cup with silicone molding. The cup's top center is attached to the switch part of the BM (Figure 2B). During perching, the impact between the suction cup and the surface changes the BM's state and flattens the suction cup, making the cup attach to the surface. After perching, since the rigid base of the BM is connected to an aerial robot, the force only applies to the cup's center (instead of the edges), thereby generating a strong holding force. To ensure the BM can only move linearly during its switching of state, we add a linear guide that can only move up and down along a hole in the base.

A self-locking mechanism is designed to lock the BM instantly upon switching its state to the collapsed state. This is realized by a passive spring-loaded rotating piece and an electromagnet for unlocking (Figure 2C). As the BM transitions from the extended to the collapsed stable state, the linear guide moves upward with an end piece pushing the rotating piece outward to rotate counterclockwise (top of Figure 2C). Once the end piece passes a notched segment of the rotating piece, the spring pushes the rotating piece back to its original position, and the notch prevents the end piece from moving downwards (bottom of Figure 2C). The electromagnet can be activated to pull the rotating piece back, releasing the end piece and unlocking the mechanism. Further details on the locking mechanism are provided in the design section.

We leverage two types of artificial muscles, both made of shape memory alloy (SMA), for tuning the perching mechanism to adapt with different contact speeds and orientation misalignment. A few works on perching have utilized SMA actuators previously.[42–44] While some studies[42,43] use SMA actuators directly for perching, our approach focuses on using SMAs to tune the mechanism's triggering force and suction cup angle. The SMAs also help in detachment; however, their main role is active tuning. In contrast, the work in ref. [44] uses SMAs for releasing spines after perching rather than for tuning or attachment.

In this study, we first tune the triggering force for the BM using SMA wires, which we call bistable mechanism actuators (BMAs) in this article. These SMA wires are wrapped around the two beams of the BM (Figure 2D). If voltages are applied to the BMAs, horizontal forces will be applied to the beams, increasing the BM's triggering force that can absorb the impact associated with increasing speeds. Second, we use four SMA springs affixed to the edges of the suction cup (Figure 2D) to tune the alignment between the suction cup and the surface. We call these SMA springs suction cup actuators (SCAs). We can actuate any two of the four SCAs to pull the suction cup and accommodate misalignments by making the suction cup roughly parallel to the surface. We opt for SMA wires for BMAs as they can generate large forces with minimal displacement, suited for tuning the triggering force. We choose the SMA springs for SCAs because they provide ample displacements with moderate forces required to pull the suction cup.

2.2. Perching Procedure

With the working principle for the whole mechanism, we explain a cycle of the perching procedure with a simplified sketch (Figure 2E). Without loss of generality, we consider the case that the whole mechanism is used to let an aerial robot perch onto a tilted surface (Figure 2E) for easy understanding, which can be extended to surfaces with other orientations. During a cycle of perching and detaching, the mechanism can undergo the following states: an initial unattached state S_u, a tuning state S_t, an attached state S_a, a detaching state S_d, and two intermediate states S_{ai} and S_{di}. In the following, we discuss the motion cycle using transitions between these states.

2.2.1. Process P_{ut} (from S_u to S_t)

The mechanism is at the initial unattached state S_u. In this state, the BM is in the extended stable state. The BMAs and the SCAs are relaxed. The mechanism is at its default triggering force (F_t) and the suction cup is not pulled at all. The rotating piece of the locking mechanism is at its default position. Before perching, we can either tune the mechanism or skip the tuning if unnecessary. With tuning, the mechanism transits from S_u to the tuning state S_t by applying a voltage to the BMAs or SCAs to increase the triggering force (F_t) to deal with large landing impacts or pull the suction cup to handle large orientation misalignment, respectively.

2.2.2. Process P_{ta} (from S_t to S_a via S_{ai})

After proper tuning at state S_t, the mechanism can perch onto a given surface, transiting to the attached state S_a through the intermediate state S_{ai}. During this process, the soft cup will contact the surface to generate an impact force to push the BM to switch the state. As the BM switches, the end piece on the linear guide will push the rotating piece of the locking mechanism through the intermediate state S_{ai}. Eventually, the BM will switch to the collapsed stable state, and the locking mechanism will fully engage. The mechanism is now attached to the surface and arrives at the attachment state S_a. Both BMAs and SCAs will be powered off to become relaxed. When the mechanism is in S_a, it can hold a large force, ensuring the mechanism can stay attached since the detaching force will come from the cup's center if the mechanism is attached to an aerial robot.

2.2.3. Process P_{au} (from S_a to S_u) via S_d and S_{di}

To detach from the surface and return to the initial unattached state S_u, we first energize the electromagnet to pull the locking mechanism's rotating piece to unlock the linear guide, thereby freeing the BM. All four SCAs are also activated at the same time. This is the start of the detachment state S_d (Figure 2E). After that, we apply a pulling force to the mechanism's base (this force can be generated from a drone's thrust). During this process, since the soft cup is initially still attached to the surface, the pulling force is transferred to the BM to switch its state. In other words, this pulling force will move the base, generating a relative motion between the linear guide and the base. As the base moves and the BM switches to the extended stable state, the SCAs are tensioned again and start to pull the edges of the suction cup (S_{di} in Figure 2E). This causes the air to leak into the cup and release the cup from the surface. After the detachment, the mechanism

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now returns to the initial unattached state S_u. The whole process forms a loop and allows the mechanism to successfully perch onto a surface in a repeated manner.

3. Design and Fabrication

In this section, we elaborate on the design and fabrication of the tunable perching mechanism with the whole assembly (**Figure 3**A) and the four mechanisms: the self-locking mechanism, the BM, the bistable mechanism actuator (BMA), and the suction cup with the suction cup actuator (SCA), which are individually shown in Figure 3B–E, respectively.

The self-locking mechanism (Figure 3B) includes a passive spring-loaded rotating piece and an electromagnet. The housing for the locking mechanism is an extension of the base on the top. The rotating piece can lock the mechanism using a small notch on top that stops the end piece once the end piece moves on top of the rotating piece. A small metal plate is attached to the rotating piece. When the electromagnet is activated, this metal plate pulls the rotating piece with it and allows the end piece to move again, unlocking the mechanism.

The BM (Figure 3C) includes one switch part, two flexible beams, two beam heads inserted at the beam's end, two rigid links to connect the switch part with beam heads, and the rigid base with a hole for the linear guide rod, which can ensure the rod's linear motion. The two rigid links have a revolute joint at both ends to connect to the switch part and the beam head. All of them except the beams and the linear guide are fabricated through Fused Deposition Modeling (FDM) 3D printing. The two flexible beams are flat carbon fiber (width × thickness: 7.9 × 0.8 mm, CF312032048, Goodwinds Composites LLC). The linear guide is a carbon fiber rod with a diameter of 2 mm.

Tuning of the BM's triggering force is realized by using a pair of SMA wires (Diameter: 0.25 mm, 90 °C, Dynalloy Inc.) wrapped around the two beam heads (Figure 3D). In this way, the BMAs surround the BM to avoid interfering with the moving components in the mechanism. To fix the BMAs in place, two plates are attached to the beam heads. The plates have metal bits that route the SMA wires and protect the parts from heat damage. The beams need to flex outward for the BM to switch state. When a voltage is applied to the BMAs, they resist this outward movement of the beams, increasing the triggering force (F_t).

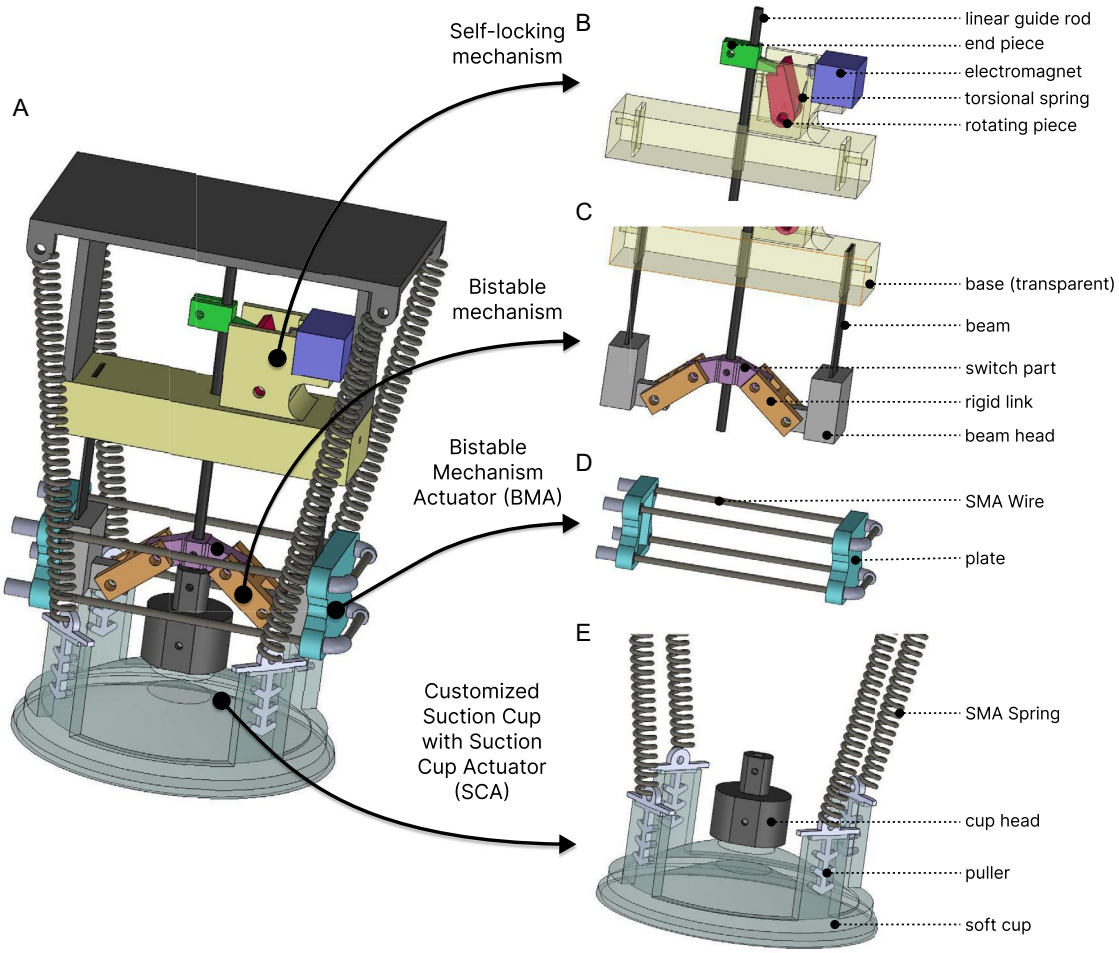


Figure 3. The design of the perching mechanism, which includes four major parts. A) The whole mechanism; B) the self-locking mechanism; C) the bistable mechanism; D) the bistable mechanism actuator (BMA); E) the customized suction cup with suction cup actuators (SCA).

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Details for how the triggering force changes with respect to voltages applied to the BMAs can be found in Section 4.1.

The suction cup (Figure 3E) is mainly made of a customized soft silicone cup connected to the BM's linear guide rod through a 3D-printed cup head. Four 3D-printed pullers are embedded inside the cup's edges. These pullers are attached to the SCAs for tuning and easy detachment from surfaces after perching. The soft cup is fabricated using the molding approach using Ecoflex 30 (the detailed process can be found in our previous article[38]).

Tuning of the suction cup's alignment is realized with four SMA Springs (SCAs) (Outer Diameter: 3.45 mm, Wire Diameter: 0.51 mm, 90 °C, 40 coils; Dynalloy Inc.) as shown in Figure 3E. One end of each SCA is attached to the suction cup edges and the other end to the base. Initially, SCAs stay relaxed and do not influence the suction cup. When a voltage is applied to two neighboring SCAs, they contract and pull the suction cup in one direction. We can control the cup's bending direction and magnitude by choosing which two SCAs to activate and the voltage, respectively. We can also activate all SCAs to pull the suction cup for detachment. Details for how SCAs will bend the cup with respect to the voltages applied to SCAs can also be found in Section 4.1.

4. Results

In this section, we discuss the characterization of the SMA actuators and show how tuning the mechanism with them can increase the success rate for perching under different contact speeds and orientation misalignment.

4.1. Characterizations of the Tuning Actuators Using Slow Steady-State Heating

The first experiment aims to evaluate the how actuating BMAs will influence the triggering force for the BM. We first measure the forces generated by the BMAs under different constant voltages after reaching a steady state. The experimental setup can be seen in Figure S1, Supporting Information, with detailed procedure in Note S1, Supporting Information. The steady-state force with respect to five different voltages (0, 2.4, 2.6, 2.8, and 3 V) are plotted in **Figure 4**A. The graph shows the average steady-state force, alongside the range of minimum and maximum steady-state force across four trials.

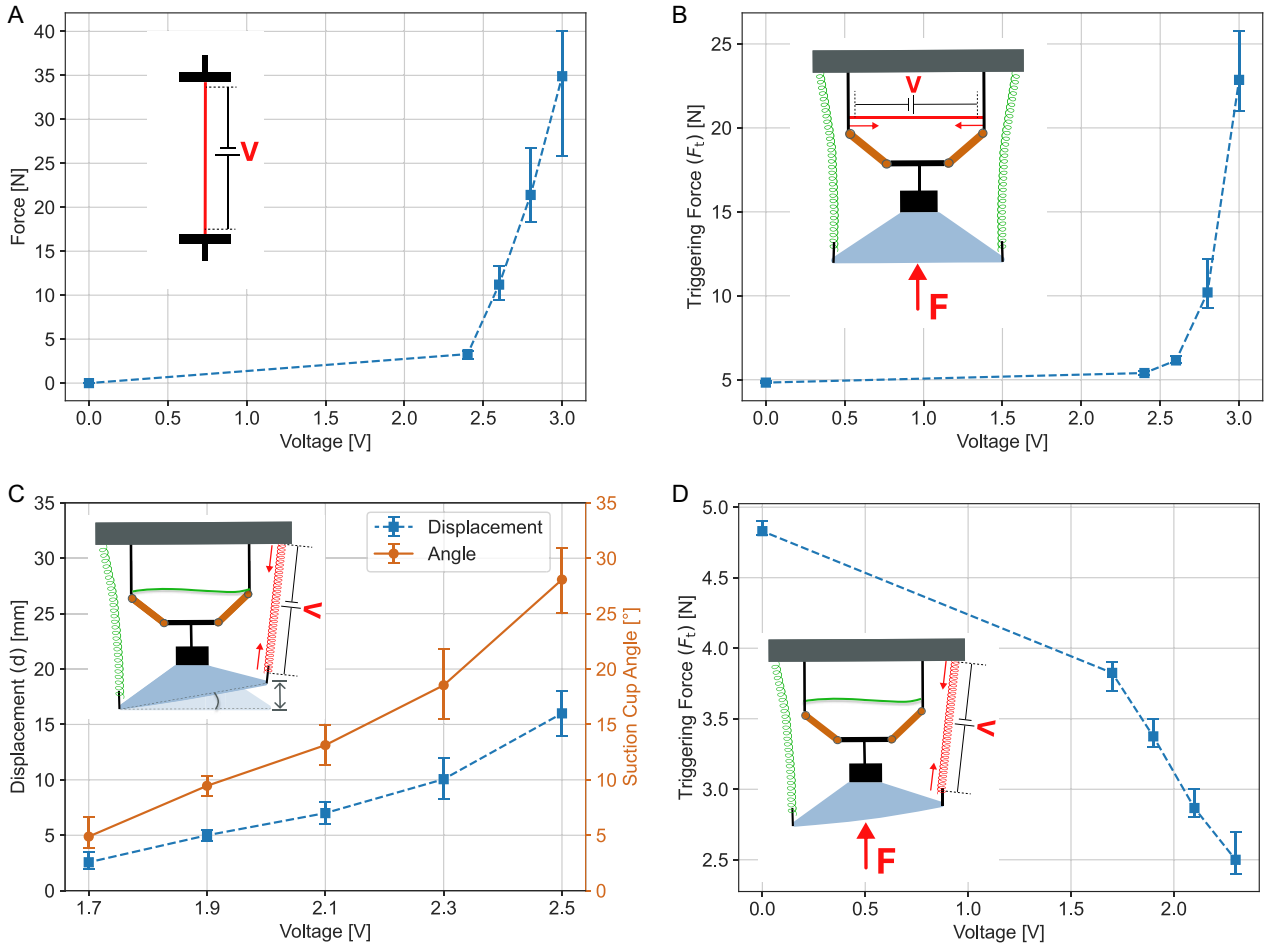


Figure 4. Experimental results for characterizing the tuning capability. A) Steady-state force versus voltages applied to BMAs; B) triggering force for the BM versus voltages applied to BMAs; C) displacement/angles for the soft cup versus voltages applied to the SCAs; D) triggering force for the BM vs voltages applied to SCAs.

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We can observe that the force increases sharply with increasing voltage, up to a very large value: >30 N at 3 V. However, under conditions where the BMAs are allowed to contract (e.g., when used in the BM where both ends are not fixed), the force is expected to be lower. This can be observed in the results of the next experiment, triggering force for the BM when BMAs are actuated.

BMAs, when activated, can increase the BM's triggering force (F_t). The second experiment is to test the required triggering force (F_t) to switch the BM state with different voltages applied to the BMAs. We conduct this experiment with the whole mechanism, i.e., with the suction cup and suction cup actuators (SCAs) attached to the mechanism (detailed experimental procedure can be found in the Note S1, Supporting Information). The force versus time graph for one of these experiments is shown in Figure S3, Supporting Information. From Figure S3, Supporting Information, we see that the force will monotonically increase to a maximum value and then rapidly drop to zero. This maximum force is the "triggering force (F_t)" required for switching the BM. We plot the triggering force at different voltages in Figure 4B. Similar to the first experiment, we also present the average triggering force, along with the range of minimum and maximum triggering forces across four trials. By applying voltage to the BMAs, the triggering force can be increased substantially: from ≈5 N when the BMAs are not actuated to ≈25 N when 3 V is applied. We can get intermediate triggering forces by applying different voltages to the BMAs.

The next set of experiments is for the SMA springs or the suction cup actuators (SCAs). The purpose of using the SCAs for tuning is to pull the suction cup in one direction to align with the surface when a misalignment exists. Thus, we perform an experiment to activate two adjacent SCAs to measure the amount of displacement of the suction cup with different voltages applied. From this displacement, we also calculate the angle of the suction cup at different SCA voltages. The graph of suction cup displacement/angle versus SCA voltages is plotted in Figure 4C. From this figure, we can see as the voltage increases, the suction cup displaces/rotates more. However, the voltage applied to the SCAs cannot be too large since the BM may be inadvertently triggered to switch state under a large force from the SCAs.

Finally, we conduct an additional experiment to assess the triggering force when activating any two of the SCAs, using a setup similar to the previous triggering force experiment. Only two of the four SCAs are activated since we only use SCAs for tuning the suction cup, as previously discussed. This investigation is essential since when the SCAs pull the suction cup, the pulling force also acts on the BM to decrease the triggering force. The resulting graph of this experiment is shown in Figure 4D. From the figure, we see that the triggering force decreases with increasing voltage applied to the SCAs. This result suggests that as aligning the suction cup becomes more critical during perching with larger misalignments, leading to a decreased triggering force, we may need to compensate for the diminished triggering force by increasing the voltage applied to the BMAs, as a higher voltage to the BMAs increases the triggering force.

4.2. Fast Tuning Using Momentary High Voltage Heating

For rapid perching and energy efficient activation of SMA actuators, fast tuning is essential. In this study, we explore the fast-tuning capabilities of both SMA wires (or BMAs) and SMA springs (or SCAs) to achieve target force or displacement quickly with minimized energy expenditure.

SMA wires are used for force generation. They exhibit rapid response times when subjected to high-voltage activation. In this case, we can actuate SMA wires using a high voltage for a short amount of time to reach a desired force level and then use a low voltage to maintain that force. This open loop strategy is experimentally investigated and the results are shown in **Figure 5**. From the figure, the force produced by the SMA wire increases rapidly upon the application of a 10 V pulse. This high voltage phase (t_1) enables the wire to reach the target force level within 100–160 ms, depending on the desired output. Following this initial phase, there is a short delay ($t_2 = 30$–60 ms) with no voltage applied to help steady holding of the force in t_3 phase. In t_3 phase, a lower holding voltage (e.g., 4 V) is applied for $t_3 = 2$ s to maintain the generated force. This would allow us to tune the mechanism 2 s before perching. The holding phase (t_3) can be extended, allowing more time between tuning and perching. However, it is not required for rapid perching as it takes <100 ms for perching (attachment) with the current mechanism. The three distinct phases- t_1 (high voltage), t_2 (delay), and t_3 (holding voltage)—allow generating a sustained force with low energy consumption. This capability to maintain force is evident in Figure 5A, which shows the case where we want to generate a desired force of 25 N held steadily during the t_3 phase, with the true force to be 24.5 ± 1.2 N. Figure S6, Supporting Information, demonstrates generating different force levels across a range of 7–35 N, without showing specific time intervals. The ability to adjust the high-voltage pulse and its duration (t_1) along with the holding voltage enables relatively precise control over the force output.

Figure 5B shows different average forces generated and their ranges across 3 trials with fast tuning of the SMA wires. From the figure, fast tuning of SMA wires can generate similar levels of forces as with the steady state heating in Figure 4A. Furthermore, the maximum spread of the force across 3 trials using the fast tuning approach is 4.5 N at a target force of 35 N, compared to 14 N under steady-state conditions.

A similar strategy can be applied to SMA springs (or SCAs) to change the suction cup's alignment and maintain its position. However, SMA springs do not require a holding voltage during this period because their displacement changes slowly once the voltage is removed. Within the short duration (<2 s), the change in displacement is observed to be minimal, although they gradually go back to their original position over longer periods (≈5 min). Figure 5C shows the average displacement of the SMA springs and their range across 3 trials using high voltage (15 V) in fast tuning mode. The figure indicates that the SMA springs can displace rapidly with a short application of high voltage. The secondary *Y*-axis in this plot displays the calculated suction cup angle. We can see that the fast tuning method for SMA springs can generate similar level of displacement and angle as of the steady state heating shown in Figure 4C.

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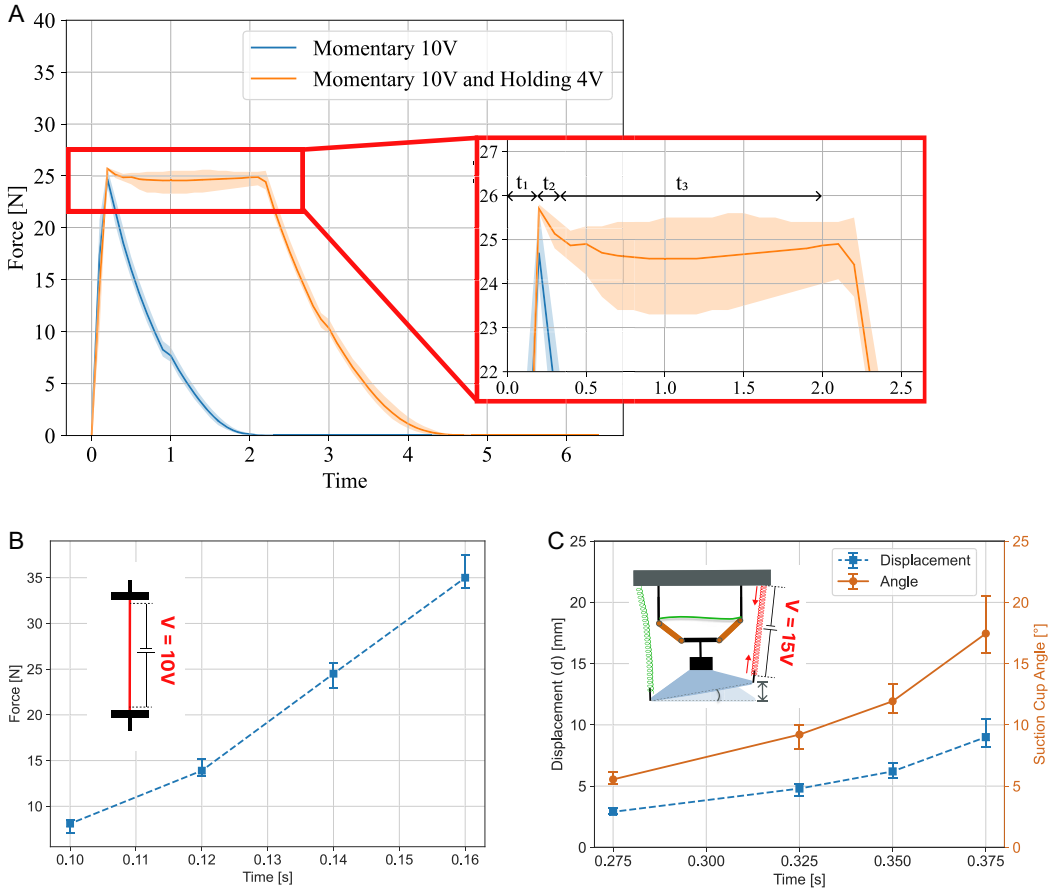


Figure 5. Experimental results demonstrating the fast tuning capability for SMA actuators. A) Fast tuning stages for SMA wire (or BMAs); B) force versus time for SMA wire; C) displacement versus time for SMA spring (or SCAs).

We compare the energy efficiency of the SMA actuators between steady-state and fast tuning modes, with results summarized in Note S3 and Figure S7, Supporting Information. Fast tuning significantly reduces energy consumption for both SMA wires and springs by minimizing heat dissipation during brief high-voltage activation, highlighting its suitability for rapid and efficient tuning in perching applications.

4.3. Perching Performance for the Tunable Mechanism Under Different Scenarios

This section explores how the tunable mechanism influences perching success rates under various contact speeds and orientation misalignments. Through systematic experiments, our objective is to evaluate the effectiveness of the mechanism in improving the results of perching in three different scenarios: 1) pendulum-based perching; 2) ground perching/landing; and 3) ceiling perching (powered flight perching). First, we attach the mechanism to the end of a pendulum to simulate varying contact speeds and landing impacts in a controlled manner. We then extend our evaluation to ground-based perching, where the mechanism is mounted on a drone (quadrotor) and dropped onto a misaligned surface, simulating uncontrolled landing scenarios. Lastly, we assess ceiling perching by flying the drone equipped with the mechanism to examine the perching performance at various angles. All three experimental setups are depicted in detail in Figure S8, S10, and S11, Supporting Information, respectively. These scenarios provide a comprehensive understanding of the mechanism's performance across a range of real-world conditions. In the following subsections, we elaborate each perching experiment scenario. We also compare the results in each scenario and examine the factors influencing their success rates.

4.3.1. Pendulum-Based Perching

As our first step, we attach the mechanism to the end of a pendulum for its simplicity to generate different contact speeds and landing impacts (a detailed experimental setup is depicted in the Figure S8, Supporting Information). This setup includes a pendulum utilizing a joint with three rotational degrees of freedom (DOFs) as detailed in Figure S9, Supporting Information. The three rotational DOFs make the pendulum setup more versatile than a standard 1D pendulum.

While not fully unconstrained like the dynamics of a free-flying quadrotor, the pendulum setup still provides meaningful

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insights by simulating certain real-world scenarios. For instance, inverted landing, as described in ref. [17], involves a quadrotor making initial attachment with its legs, followed by a pivoting motion about the attachment point driven by momentum. Our pendulum setup parallels this behavior, as the attachment point of the pendulum results in a similar pivoting motion. Under this simplified experimental setup, we compare perching success rates with and without tuning to evaluate the tuning mechanism.

First, we conduct the experiment at three different contact speeds without any misalignment. The different speeds are realized by releasing the pendulum at three different angles: 15°, 30°, and 45°, which correspond to 0.74, 1.46, and 2.16 m s^{-1}, respectively. Then we conduct experiments with different misalignments at each of the three speeds. For all the experiments, we first start with no tuning, i.e., 0 V applied to both BMAs and SCAs. Then we tune the mechanism by applying different voltages to the BMAs and the SCAs and continue the experiments for different cases. We use the steady-state heating method for its simplicity. Specifically, voltage is applied to the BMAs for 60 s and to the SCAs for ≈2 min to allow them to reach a steady state. We waited sufficiently and made sure they were in steady state before conducting each trial. In real perching scenarios, we can always use fast tuning to ensure rapid perching to minimize the energy consumption. The combined results of all the experiments are shown in the success matrix in **Figure 6**. For all experiments, 4 trials are conducted to ensure repeatability.

For experiments involving misalignments ranging from 0° to 17.5°, all cases result in successful perching across the three speeds tested, even without any tuning, indicating the suction cup has some capability to passively handle misalignment due to its softness. Therefore, we opt not to include these results in the success matrix figure (Figure 6) to prioritize interesting and important results. The experiment results for misalignments ranging from 20° to 30°, conducted at all three speeds, are visualized in Figure 6. A notable trend can be seen from the figure: as the misalignment angle increases, the success rate declines to 0% without tuning. Tuning substantially enhances perching success, especially at larger misalignment angles. For smaller misalignment angles, applying voltage only to the BMAs proves to be effective. This is because the suction cup flattens easily with sufficient resistance (sufficient triggering force) and adheres to the surface. However, as the misalignment angle increases, simply actuating BMAs is not enough, voltage also needs to be applied to the SCAs to align the suction cup better with the surface. This, in turn, requires a higher BMA voltage to keep the triggering force sufficient. Consequently, both BMA and SCA voltages must increase for successful perching at increasing misalignment

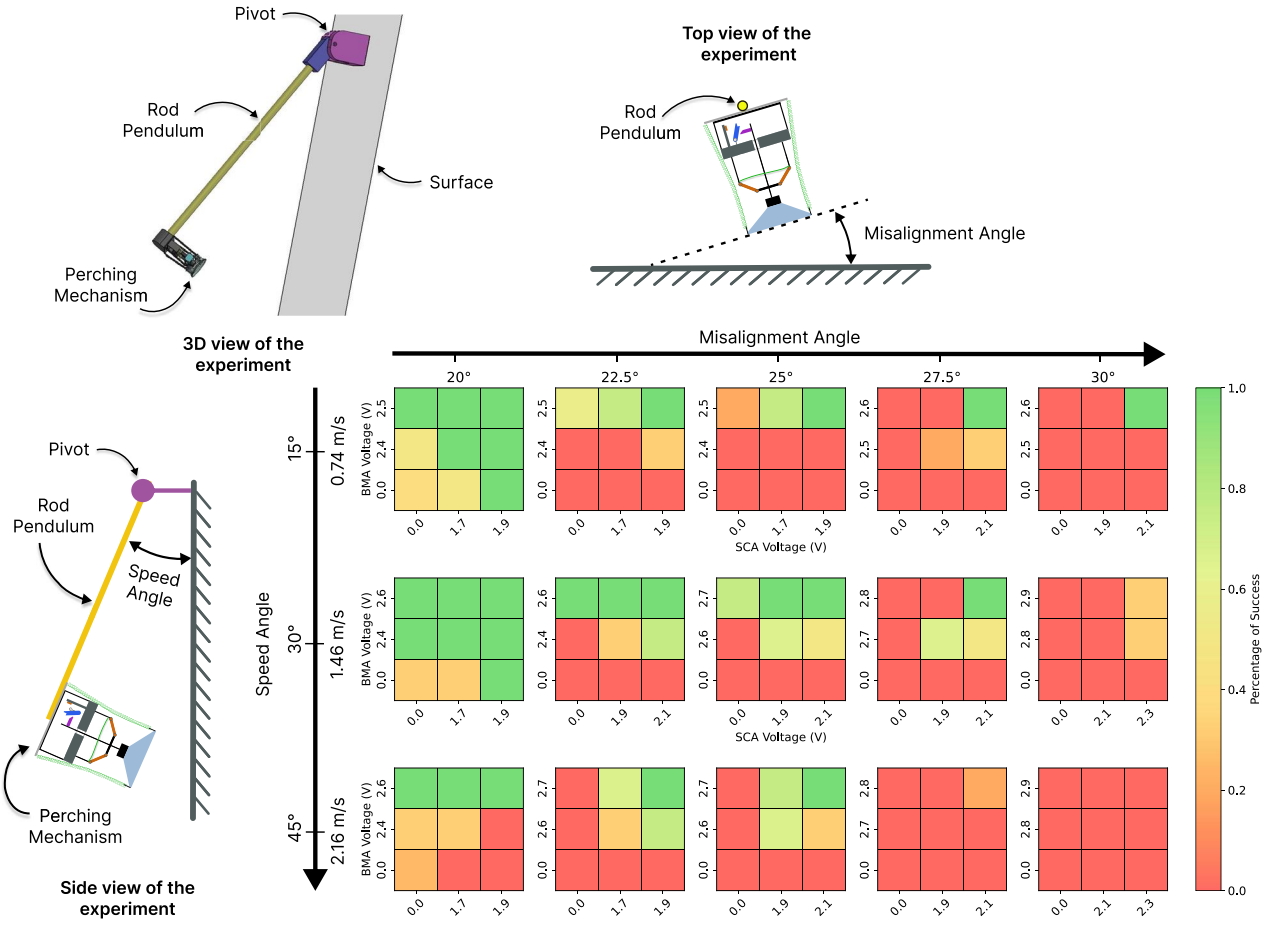


Figure 6. Experimental setup and results for pendulum-based perching using the tuning capability of the mechanism.

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angles. This need intensifies at higher speeds, as perching success diminishes with increasing speed at higher misalignment angles.

We can further correlate the suction cup angle and perching success by examining the suction cup angle from Figure 4C and the success matrix in Figure 6. The suction cup appears to successfully accommodate and attach when the surface misalignment is on average within 15° after tuning. This accommodation tends to increase at lower speeds but decreases as speeds rise. At very large misalignments (e.g., >32.5°), no amount of tuning seems effective, as the moment rotating the mechanism (A2 in Figure S12 and S13, Supporting Information) becomes too significant, while the normal force components flattening the suction cup becomes smaller.

To further understand how the mechanism behaves during the perching process, we recorded multiple failed and successful perching for some contact speeds and misalignment angles. **Figure 7** presents a sequence of snapshots for different perching attempts from a bottom view (detailed videos can be found in Movie S1, Supporting Information). The sequences in Figure 7 include seven images numbered from left to right. Image 1 depicts the "Approach" step, where the mechanism

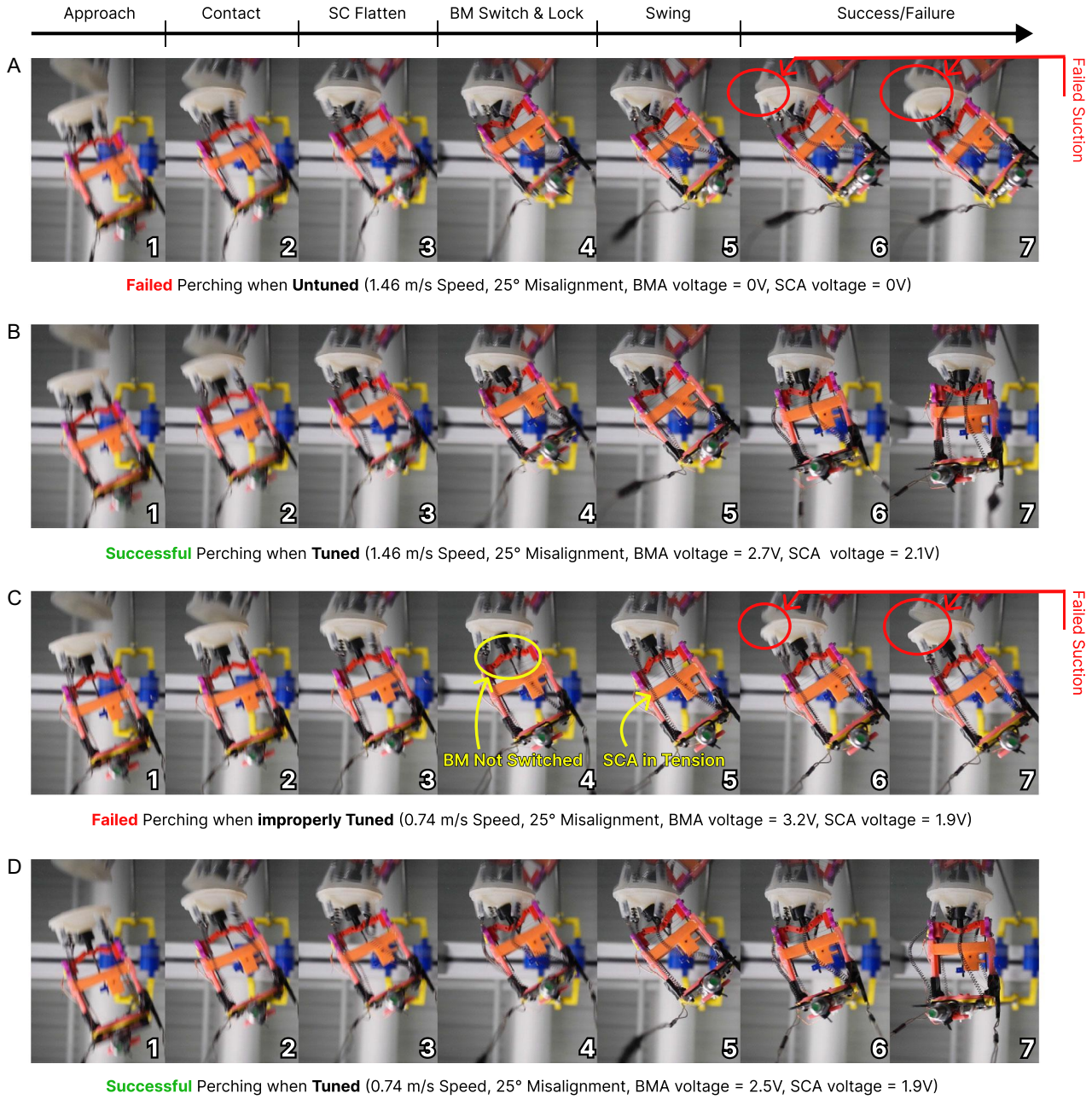


Figure 7. Experimental results for pendulum-based perching using the tuning capability of the mechanism. A,B): Failed and successful perching for 1.46 m s^{-1} contact speed and 25° misalignment, respectively. C,D): Failed and successful perching for 0.74 m s^{-1} contact speed and 25° misalignment, respectively.

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is nearing the surface but has not yet made contact. Image 2 represents the moment of contact. Image 3 shows the suction cup flattening and the BM in an intermediate state. Image 4 depicts the BM fully switched with the locking mechanism engaged. Image 5 illustrates the continuation of the swing due to misalignment-induced rotating moment. Images 6 and 7 show the final stages of the perching, which may result in either successful perching or failure.

Four perching attempts are displayed in Figure 7, showing both untuned and differently-tuned scenarios. Sequence A (Figure 7A) shows a failed attempt at a speed of $1.46\,m\,s^{-1}$ and a misalignment of 25° without tuning, where the suction cup begins to detach from the left edge in images 6 and 7. In sequence B (Figure 7B), successful perching is achieved under the same conditions but with proper tuning.

Subsequent sequences (Figure 7C,D) underscore the importance of BM switching. Both sequences feature a perching speed of $0.74\,m\,s^{-1}$ and a 25° misalignment, but with different tuning. In sequence C, the BMA voltage is set too high at 3.2 V for the speed of $0.74\,m\,s^{-1}$, which prevents the BM switch, leading to failed perching as seen in images 6 and 7. Since the BM remains in the extended state, the SCAs are in tension (Image 5) as the mechanism pivots around the suction cup due to the misalignment-induced moment. In the 6th image, we observe the suction cup beginning to peel from the surface at that edge, resulting in failed perching. Conversely, in sequence D, proper BMA tuning at 2.5 V results in the expected BM switch, enabling successful perching. These results highlight the critical role of BM switching in perching success. Additionally, failure to switch can lead to increased moment arm length (mechanism length longer) and potentially failed perching due to excessive rotating moment generation. The failure modes along with the moments influencing the perching process are explained in detail in Figure S12 and S13, Supporting Information.

4.3.2. Ground Perching or Free-Fall Landing

In this scenario, the mechanism is mounted on a drone (quadrotor) and dropped onto a glass surface with varying misalignment angles (30°, 35°, and 40°). The misalignment was created by raising two legs of a glass table, tilting its surface at the desired angle. A tether was used to drop the drone from a certain height. The experimental setup is shown in Figure S10, Supporting Information. The drop heights were set to 10 and 20 cm to generate different impact velocities. The corresponding velocities for 10 and 20 cm drop are 1.4 and $1.98\,m\,s^{-1}$, respectively. We performed the experiments with untuned and tuned mechanism similar to the pendulum case to compare the success rates of both cases. These experiments aimed to evaluate the tunable mechanism's ability to perch under uncontrolled or unpowered conditions, where the drone does not actively contribute to the perching process after release.

The results for the ground perching experiments are presented in **Figure 8**A. The results demonstrate that the untuned mechanism is only successful in perching under the 30° misalignment and $1.4\,m\,s^{-1}$ speed, achieving 40% success across five trials. In contrast, the tuned mechanism consistently achieves better perching success across all conditions tested, including higher misalignment angles and drop velocities. The success rate for the tuned mechanism for all conditions is 100% across five trials except at 40° misalignment and $1.98\,m\,s^{-1}$ speed, where it is 80% successful. In Figure 8, we tune both the BM and the suction cup with BMA voltage = 2.7–3.1 V and SCA voltage = 2.1–2.3 V. The ability to align the suction cup with the surface and increasing the triggering force for the BM proved crucial in accommodating both higher misalignments and increased impact velocities to ensure successful perching outcomes.

Figure 8B,C depicts the image sequence of one typical failed and successful perching, respectively. Specifically, Figure 8B shows failed perching at $1.4\,m\,s^{-1}$ speed and 35° misalignment when untuned, whereas Figure 8C presents successful perching at the same condition with a tuned mechanism. Images 1–3 illustrate the sequence of events: initial condition before dropping, contact and suction cup flattening, followed by the switch of the BM which engages the locking mechanism. Image 4 captures the rebound swing caused by the collision between the drone and the surface. Image 5 shows the continuation of this swing due to gravity, leading to the final perching outcome depicted in Image 6. The tuned mechanism ensures a secure attachment capable of withstanding the continued swing, whereas the untuned mechanism experiences suction failure or detachment due to the swing. The ground perching sequence and failure modes are explained in detail in Note S5 and Figure S12 and S13, Supporting Information.

4.3.3. Ceiling Perching or Powered Flight Perching

For this case, the glass table was fixed at various misalignment angles (25°, 30°, 40°, and 45°) relative to the floor, similar to the ground perching experiments. The drone, equipped with the perching mechanism, was manually flown from underneath the table to perch onto the glass surface. The drone was configured to fly level with the horizon to ensure a consistent misalignment angle upon contact. The experimental setup is shown in **Figure 9**A.

Ideally, we should precisely control the contact speed for comparison; however, our current setup does not allow us to do so since we can only manually control the quadrotor to fly upwards without controlling the speed. Although this is nonideal, we think it still provides a qualitative comparison between the untuned and tuned case.

The mechanism proved highly effective for perching, even at very high misalignment angles (45°), as the thrust from the drone's propellers would actively push the suction cup against the surface after initial contact, helping to flatten it. At 40° misalignment angle, 22 trials are performed for both tuned (BMA voltage 3 V, SCA voltage 2.3 V) and untuned mechanisms without precisely controlling the drone's speed. The untuned mechanism achieves a success rate of 54.5%, while the tuned mechanism succeeds in 90.9% of trials, demonstrating a substantial improvement. Since the contact speed is not controlled, we qualitatively observe that the tuned mechanism generally achieves successful perching across both lower and higher velocities while the untuned mechanism struggles to flatten the suction cup at lower velocities, resulting in a higher failure rate. By

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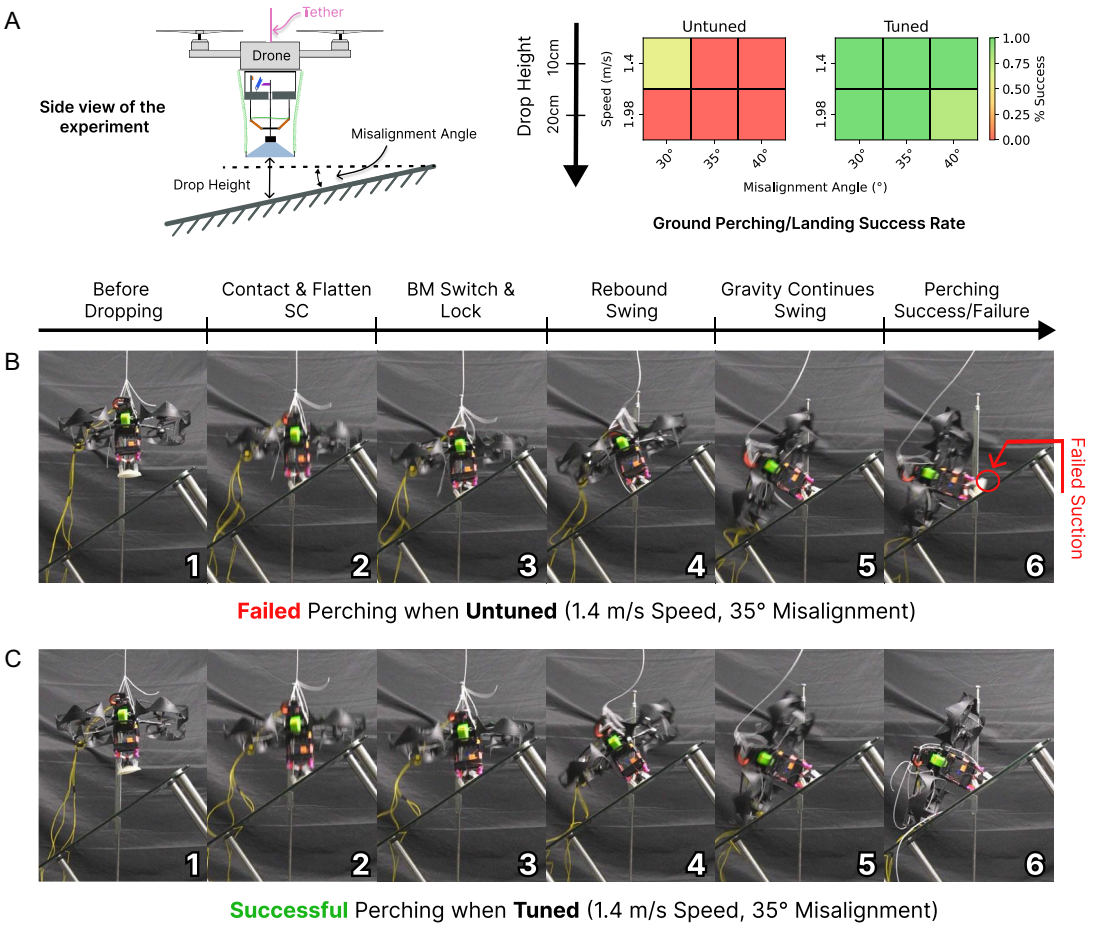


Figure 8. Experimental results for ground perching or free-fall landing. A) Experimental setup schematic and success matrix; B) failed perching 1.4 m s^{-1} contact speed and 35° misalignment when untuned; C) successful perching at 1.4 m s^{-1} contact speed and 35° misalignment when tuned.

aligning the suction cup with the surface and stiffening the BM, the tuned system ensures effective suction cup flattening under varied conditions. Conversely, the untuned mechanism often depends on higher velocity or the active push of the drone after contact to achieve sufficient flattening of the suction cup, particularly at larger misalignment angles.

Figure 9B–D shows images of three typical ceiling perching demonstration. Figure 9B shows a failed perching attempt at 40° misalignment with the untuned mechanism, while Figure 9C,D shows successful perching at 40° and 45°, respectively, with the tuned mechanism. Note that the untuned mechanism could also achieve perching success at 45° misalignment, provided sufficient velocity or an active push from the drone. The perching sequence consists of seven images where Images 1–4 is the same for all scenarios: approach, initial contact, suction cup flatten, and BM switch and lock. The distinction between the untuned and tuned mechanisms is clearly visible in Image 2 and Image 3. In Image 2, the suction cup in the tuned cases appear more flattened upon contact, as it is better aligned prior to impact. In Image 3, the untuned mechanism shows the BM in a horizontal position, indicating less resistance for the suction cup to flatten. Conversely, in the tuned cases shown in Image 3, the BM provides greater resistance to the change of state, resulting in improved flattening of the suction cup. Image 5 captures a rebound swing caused primarily by the drone's collision with the surface, similar to the ground perching case. However, in ceiling perching, gravity counteracts the rebound swing and tries to restore equilibrium. This is depicted in image 6. This can lead to some back-and-forth oscillations between the rebound swing and the gravitational correction, unlike the ground perching where both forces contribute to swings in the same direction. This back swing generated by the gravity leads to the success or failure of the perching as seen in image 7. The failure modes and the moments that the mechanism experiences are explained in detail in Note S5 and Figure S12 and S13, Supporting Information.

These experiments demonstrate the robustness of the tunable perching mechanism in powered perching scenarios. While higher velocity or the thrust provided by the drone can compensate for some limitations in untuned mechanisms, the tuned mechanism's ability to adapt dynamically to both misalignment and velocity enhances its success rate significantly.

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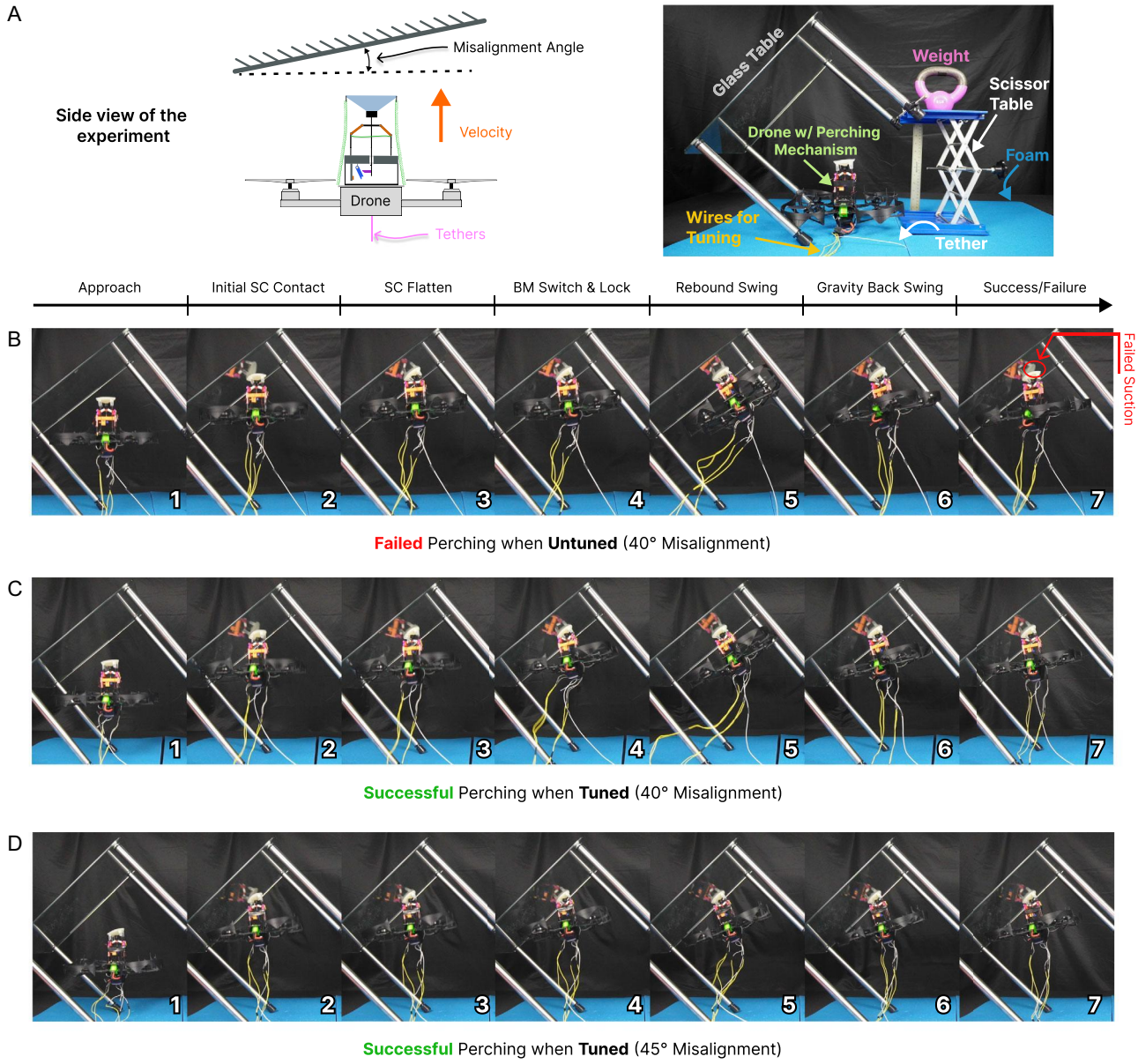


Figure 9. Experimental results for ceiling perching or powered flight perching. A) Experimental setup schematic and image; B) failed perching at 40° misalignment when untuned; C) successful perching at 40° misalignment when tuned; D) successful perching at 45° misalignment when tuned.

4.3.4. Comparison Between the Perching Scenarios and Perching Success

The three perching scenarios—pendulum-based, ground perching, and ceiling perching—demonstrate varying success rates due to differences in perching dynamics and external factors. The pendulum-based and ground perching scenarios represent unpowered cases where the drone or pendulum does not actively contribute to the perching process after the initial motion. These setups rely solely on the impact dynamics and the tunable mechanism's properties to achieve perching. In contrast, ceiling perching involves powered perching, where the drone actively uses its thrust to assist the perching process, enhancing success even under challenging conditions. Additionally, different moments associated with each scenarios determine the failure points and greatly affect success rates. Gravity plays a crucial role in ground and ceiling perching.

Pendulum-based perching shows the lowest success rate, accommodating up to 30° of misalignment at 0.74 m s^{-1} due to limited resistance against the rotating moment and its higher mass (850 g). Ground perching achieves intermediate success, handling up to 40° misalignment at 1.98 m s^{-1}, despite destabilizing gravitational effects. Ceiling perching demonstrates the highest success rate, managing misalignments as high as 45° across various speeds due to the active push from the drone and gravity's stabilizing influence. The comparison between

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the three perching scenarios and how they fail is explained in detail in Note S5 and Figure S12 and S13, Supporting Information.

5. Discussion and Conclusion

In this study, we developed a novel tunable mechanism for aerial robots to enable robust perching onto smooth surfaces under different conditions. The tuning capability of our mechanism allows it to adapt to a wide range of controlled contact speeds (up to 2.16 m s^{-1}) and orientation misalignments (up to 45°). The mechanism could potentially adapt to higher speeds, as observed in the ceiling perching experiments, demonstrating its robustness beyond the conditions tested in controlled experiments. This adaptability reduces the need for precise control of speed and orientation during perching.

Experimental results across three different perching scenarios—pendulum-based perching, ground perching, and ceiling perching—demonstrate the effectiveness of the tunable mechanism in accommodating different contact speeds and orientation misalignments. Both pendulum-based perching and ground perching experiments are conducted under controlled conditions, which verified the enhanced performance of the tuned mechanism over the untuned one across various contact speed and misalignments. Additionally, the ceiling perching experiments at 40-degree misalignment across various uncontrolled speeds showed higher success rates for the tuned mechanism (90.9%) compared to the untuned mechanism (54.5%). Overall, these results underscore the significance of the tuning mechanism to enhance perching success under diverse conditions.

Moreover, our mechanism can be rapidly and reliably tuned using a momentary high-voltage pulse (<0.4 s), minimizing energy consumption (max < 50 J). The time as well as the energy consumption can be further reduced by applying higher voltages. The upgraded self-locking mechanism adds to the reliability of our mechanism, activating instantly after the BM switches without failure.

Despite the promising results obtained, our study has some limitations that suggest directions for future research. Our mechanism requires knowledge of the relative orientation between the flying robot and the surface to determine the necessary tuning adjustments. However, perching onto surfaces with unknown orientations inherently requires orientation sensing, whether to inform our tuning mechanism or to directly control a quadrotor's orientation. For surfaces with known orientations, additional sensing is not needed, as the quadrotor's onboard inertial measurement units can estimate its own orientation and, consequently, the relative orientation with the surface. This information can then guide the tuning adjustments. In essence, our tuning mechanism can operate with the same hardware setup required for perching by directly controlling a quadrotor's orientation. On top of that, for our tuning mechanism, the orientation sensing does not need to be highly precise, as the suction cup itself can accommodate minor misalignments (up to 10–15°). Similarly, the tuning adjustments needed for successful perching do not require high precision either. This allows effective perching under varied conditions without precise sensing or control.

The insights from our study suggest that the moments generated during different perching scenarios, coupled with factors such as the geometry of the drone, the mechanism, and the misalignment angles of the surface, play a significant role in determining the perching outcome. These elements influence the magnitude of the moment generated, collision of the drone with the surface, and the stability during the perching process. In addition to the tuning mechanism, the stiffness and design of the suction cup are likely factors that impact perching success, particularly in scenarios involving unpowered perching. Investigating these aspects further will provide valuable insights for improving the mechanism's adaptability and reliability across diverse operational conditions. Our future work will aim to enhance the mechanism's applicability for various perching scenarios by addressing these factors.

From our study, we conclude that a tunable mechanism can increase the success rate for autonomous perching of aerial robots without precise speed or orientation control. As long as the speed and orientation are controlled within a specific range, we can always tune the perching mechanism to ensure successful perching. This is particularly useful for challenging perching conditions such as inverted perching on the ceiling, when an aerial robot may lose its control authority in certain directions before touchdown.[17] For instance, a quadrotor may not be able to reduce its speed or actively control its orientation after flipping over for perching. Our tuning strategy also mimics the perching behavior observed in biological flies. They perform perching by actively using their legs to absorb landing impacts and orienting their legs properly before touchdown for successful perching.[45] Overall, our tunable mechanism offers a promising avenue for increasing the success rates of autonomous perching under a broad spectrum of operational conditions, paving the way for more versatile and resilient aerial robots.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

Mahmud Hasan Saikot: formal analysis (lead); writing—original draft (lead); and writing—review and editing (lead). **Feiyu Wu**: investigation (equal); validation (equal); visualization (supporting); and writing—original draft (supporting). **Elisha Lerner**: data curation (supporting); investigation (supporting); supervision (supporting); and writing—original draft (supporting). **Bo Cheng**: conceptualization (equal); funding acquisition (lead); project administration (equal); supervision (equal); and writing—review and editing (supporting). **Jianguo Zhao**: conceptualization (lead); funding acquisition (equal); investigation (lead); supervision (equal); writing—original draft (lead); and writing—review and editing (lead).

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Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Keywords

aerial perching, bistable mechanism, perching, perching mechanism, physical intelligence

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